

$$e_\mu^a(x) = \sqrt{\chi^a(x)} O_\mu^a(x)$$

$$g_{\mu\nu}(x) = e_\mu^a(x) \eta_{ab} e_\nu^b$$

$$\rightarrow \underline{M^4 \text{ Lorentz}}: x^a \rightarrow \Lambda^a_b x^b = x'$$

$$\eta_{ab} = \Lambda^c_a \eta_{cd} \Lambda^d_b$$

$$\begin{aligned} g_{\mu\nu} &= \Lambda^c_a \eta_{cd} \Lambda^d_b e_\mu^b e_\nu^a \\ &= (\Lambda^c_a e_\mu^a) \eta_{cd} (\Lambda^d_b e_\nu^b) \end{aligned}$$

$\Rightarrow$ Eq. class of frame fields related by ISO^+

$$\begin{aligned} e_\mu^a &\leftarrow \text{Lorentz} \\ e_\mu^a &\leftarrow \text{space time} \end{aligned}$$

$$e_p^{a'}(x') = \frac{\partial x^\mu}{\partial x'^p} e_\mu^a(x)$$

$$\begin{aligned} \det(g_{\mu\nu}) &= \det(e_\mu^a)^2 \det(\eta_{ab}) \\ \Rightarrow \sqrt{-\det(g_{\mu\nu})} &= \det(e_\mu^a) \end{aligned}$$

$$e \equiv \det(e_\mu^a) = \sqrt{-|g|} \neq 0$$

$$\exists e_a^\mu \Rightarrow e_b^\mu e_\mu^a = \delta_b^a$$

and

$$e_a^\mu e_\mu^b = \delta_a^b$$

$$g_{\mu\nu} = e_\mu^a \eta_{ab} e_\nu^b$$

$$g^{\sigma\tau} g_{\mu\nu} = \delta_\mu^\sigma = g^{\sigma\tau} e_\mu^a \eta_{ab} e_\tau^b$$

$$\delta^\sigma_\mu = e^\sigma_\mu (g^{\sigma\tau} \eta_{ab} e^\tau_b)$$

$$\delta^\mu_\nu = e^\mu_\nu e^\nu_a$$

$$e^\mu_a = g^{\mu\sigma} \eta_{ab} e^\sigma_b$$

$$* g_{\mu\nu} = e^\sigma_\mu \eta_{ab} e^\tau_\nu$$

$$e^\mu_c g_{\mu\nu} e^\nu_d = \delta^\sigma_c \eta_{ab} \delta^\tau_d = \eta_{cd}$$

LHS

$$(g^{\mu\sigma} \eta_{cn} e^\tau_n) g_{\mu\nu} (g^{\nu\lambda} \eta_{dm} e^\mu_\lambda)$$

$$= e^\tau_n \eta_{cn} g^{\nu\lambda} \eta_{dm} e^\mu_\lambda$$

$$e^\tau_n \eta_{cn} e^\nu_d = \delta^\tau_d \eta_{cn} = \eta_{cd}$$

* proof of "flip": in definition.

$$e^\sigma_\mu e^\mu_b = \delta^\sigma_b$$

$\Rightarrow$

$$e \rightarrow e'^a(x) = (\Lambda^{-1})^a_b(x) e^\sigma_b(x)$$

Q: Lorentz doesn't change spacetime points

$$x^\mu = \Lambda^\mu_\nu x^\nu$$

a : reps of Lorentz-Alg.

μ : reps of diffeomorphism alg.

$$e^\sigma_\mu e^\mu_b = \delta^\sigma_b \Rightarrow e^\mu_a(x') = \frac{\partial x'^\mu}{\partial x^\sigma} e^\sigma_a(x)$$

$$* e^\mu_a g_{\mu\nu} e^\nu_b = \eta_{ab}$$

$\{e^\mu_a\}$: orthonormal set of vectors in $T_p M$

• Since $\det(e_a^m) \neq 0$, then

$$T_p M = \text{Span} \{ e_a^m(p) \partial_m \}$$

↓

* The canonical basis: $V = V^m \partial_m$

$$\{ e_m \equiv \partial_m \}$$

$$\Rightarrow \text{New basis: } E_a \equiv e_a^m(x) \frac{\partial}{\partial x^m}$$

$$g(E_a, E_b) = e_a^m g(e_m, e_n) e_n^b = \eta_{ab}$$

* V is basis independent.

$$V = V^m \partial_m = V^a E_a = V^a e_a^m \partial_m$$

$$V^m = e_a^m(x) V^a(x)$$

$$V^a = e_a^m(x) dx^m$$

on
 $T_p M$

on
 $T_p^* M$

$$e^a \equiv e_a^m dx^m$$

* V^a : Transforms as $\underline{D}$ of ISO^+ but as $\underline{1}$ of $\text{Diff}_g(M)$

$$V'^a(x) = (\Lambda'^a_b)^a V^b(x)$$

$$\text{Metric: } T_p^{0,2}(M) \ni g$$

$$g = g_{mn} dx^m \otimes dx^n = \hat{e}_m^a \eta_{ab} \hat{e}_n^b dx^m \otimes dx^n = \eta_{ab} \hat{e}^a \otimes \hat{e}^b$$

$$[F, F] = [e^m, e^n]$$

$$\begin{aligned}
& [E_a, E_b]_p = [e_a^\mu \partial_\mu, e_b^\nu \partial_\nu] \\
& = e_a^\mu \partial_\mu (e_b^\nu \partial_\nu) - e_b^\nu \partial_\nu (e_a^\mu \partial_\mu) \\
& = e_a^\mu (\partial_\mu e_b^\nu) \partial_\nu + e_a^\mu e_b^\nu \partial_\mu \partial_\nu - e_b^\nu \partial_\nu e_a^\mu \partial_\mu - \dots \\
& = e_a^\mu \partial_\mu e_b^\nu \partial_\nu - e_b^\nu \partial_\nu e_a^\mu \partial_\mu \\
& \quad \mu \leftrightarrow \nu \\
& = e_a^\nu (\partial_\nu e_b^\mu) \partial_\mu - e_b^\nu (\partial_\nu e_a^\mu) \partial_\mu \\
& = (e_a^\nu \partial_\nu e_b^\mu - e_b^\nu \partial_\nu e_a^\mu) \partial_\mu
\end{aligned}$$

$$\Rightarrow F_a e_b^\mu = \partial_\mu$$

$$[E_a, E_b]_p = e_c^\mu (e_a^\nu \partial_\nu e_b^\mu - e_b^\nu \partial_\nu e_a^\mu) E_c$$

$$[E_a, E_b] = -\Omega_{ab}^c E_c$$

$$\Omega_{ab}^c = -e_c^\mu (e_a^\nu \partial_\nu e_b^\mu - e_b^\nu \partial_\nu e_a^\mu) E_c$$

Also of the form:

$$\Omega_{ab}^c = e_a^\mu e_b^\nu (\partial_\mu e_\nu^c - \partial_\nu e_\mu^c)$$

proof:

$$\begin{aligned}
& \partial_\nu (e_\mu^c e_b^\mu) = 0 \\
& \partial_\nu e_\mu^c e_b^\mu + e_\mu^c \partial_\nu e_b^\mu = 0 \\
& e_\mu^c \partial_\nu e_b^\mu = -e_b^\mu \partial_\nu e_\mu^c \\
& e_c^\sigma e_\mu^c \partial_\nu e_b^\mu = -e_c^\sigma e_b^\mu \partial_\nu e_\mu^c \\
& \partial_\nu e_b^\mu = -e_c^\mu e_b^\sigma \partial_\nu e_\sigma^c
\end{aligned}$$

$$\begin{aligned}
\Omega_{ab}^c &= -e_c^\mu (e_a^\nu [-e_b^\mu e_\sigma^c \partial_\nu e_\sigma^\mu] - e_b^\nu [-e_\sigma^\mu e_a^c \partial_\nu e_\sigma^\mu]) \\
&= \delta_a^c (e_b^\nu e_\sigma^c \partial_\nu e_\sigma^\mu - e_b^\nu e_a^c \partial_\nu e_\sigma^\mu) \\
&= (e_b^\nu e_\sigma^c \partial_\nu e_\sigma^\mu - e_b^\nu e_a^c \partial_\nu e_\sigma^\mu) \\
&= (e_a^\mu e_b^\nu \partial_\mu e_\nu^c - e_b^\nu e_a^\mu \partial_\nu e_\mu^c)
\end{aligned}$$

$$\Omega_{ab}^c = e_a^\mu e_b^\nu (\partial_\mu e_\nu^c - \partial_\nu e_\mu^c)$$

$$e^a: TM \rightarrow \mathbb{R}$$

$$\begin{aligned}
 e^a(E_b) &= e^a_\mu dx^\mu (e^b_\sigma \partial_\sigma) \\
 &= e^a_\mu e^b_\sigma dx^\mu (\partial_\sigma) = e^a_\mu e^b_\sigma \frac{\partial x^\mu}{\partial x^\sigma} \\
 &= \delta^\mu_\sigma e^a_\mu e^b_\sigma = e^a_\mu e^b_\mu = \delta^a_b \quad \checkmark
 \end{aligned}$$

ex: S^2

$$g = d\theta d\theta + \sin^2(\theta) d\phi d\phi = e^1 \otimes e^1 + e^2 \otimes e^2$$

$$e^1 = d\theta, \quad e^2 = \sin(\theta) d\phi$$

$$e^1_\theta = 1$$

$$e^1_\phi = 0$$

$$e^2_\theta = 0$$

$$e^2_\phi = \sin(\theta)$$

$$\Omega_{ab}{}^c : \Omega_{12}{}^2 = -\cot(\theta)$$

$$* \{ \gamma^a, \gamma^b \} = 2\eta^{ab} \quad \Bigg| \quad \boxed{\gamma^\mu = e^\mu_a \gamma^a}$$

$$\hookrightarrow \gamma^a = e^a_\mu \gamma^\mu$$

$$\{ \gamma^\mu, \gamma^\nu \} = e^\mu_a e^\nu_b \{ \gamma^a, \gamma^b \} = e^\mu_a e^\nu_b 2\eta^{ab} = 2g^{\mu\nu}$$

* Let $\omega^{(D)}$ be any top form:

$$\begin{aligned}
 \pm &= \int \omega^{(D)} = \frac{1}{D!} \int \omega_{\mu_1 \dots \mu_D}(x) dx^{\mu_1} \wedge \dots \wedge dx^{\mu_D} \\
 &= \int \omega_{01 \dots D-1} dx^0 dx^1 \dots dx^{D-1}
 \end{aligned}$$

$$\begin{aligned}
 \frac{1}{D!} dx^{\mu_1} \wedge \dots \wedge dx^{\mu_D} &= \delta^{\mu_1}_{\mu_1} \delta^{\mu_2}_{\mu_2} \dots \delta^{\mu_D}_{\mu_D} dx^{\mu_1} \wedge \dots \wedge dx^{\mu_D} \\
 &= \delta^{\mu_1 \dots \mu_D}_{\mu_1 \dots \mu_D} dx^{\mu_1} \wedge \dots \wedge dx^{\mu_D}
 \end{aligned}$$

$$\int \omega_{\mu_1 \dots \mu_D} \delta^{\mu_1 \dots \mu_D}_{\mu_1 \dots \mu_D} dx^{\mu_1} \wedge \dots \wedge dx^{\mu_D} = \int \omega_{01 \dots D-1} dx^0 \wedge \dots \wedge dx^{D-1}$$

(i) Form fields: $\int * F^{p+1} \wedge F^{p+1}$

(ii) Top forms: $\int \Omega^{(D)}$

ex: Chern Simons 3D gravity: $D=3$

$$S_{CS} = \frac{\kappa}{4\pi} \int A \wedge dA$$

A : $U(1)$ Gauge field, $A = A_\mu dx^\mu$

$$F = dA = \frac{1}{2} F_{\mu\nu} dx^\mu \wedge dx^\nu = \frac{1}{2} F_{\mu\nu} dx^\mu \wedge dx^\nu$$

$$dA = \frac{1}{2} \partial_\mu A_\nu dx^\mu \wedge dx^\nu = \frac{1}{2} F_{\mu\nu} dx^\mu \wedge dx^\nu$$

$$= \frac{1}{2} (\partial_\mu A_\nu - \partial_\nu A_\mu) dx^\mu \wedge dx^\nu$$

$$\bullet F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$$

$$S_{CS} = \frac{k}{4\pi} \int A_\mu dx^\mu \wedge \left(\frac{1}{2} F_{\nu\rho} dx^\nu \wedge dx^\rho \right)$$

$$= \frac{k}{8\pi} \int A_\mu F_{\nu\rho} dx^\mu \wedge dx^\nu \wedge dx^\rho$$

$$= \frac{k}{8\pi} \frac{1}{3} \int (A_\mu F_{\nu\rho} + A_\nu F_{\rho\mu} + A_\rho F_{\mu\nu}) dx^\mu \wedge dx^\nu \wedge dx^\rho$$

$$S_{CS} = \frac{k}{4\pi} \int (A_0 F_{12} + A_1 F_{20} + A_2 F_{01}) dx^0 dx^1 dx^2$$

• Coordinate invariant and also topological
 * Canonical volume form

$$dV \equiv e^0 \wedge e^1 \wedge \dots \wedge e^{D-1} = d^D x \sqrt{-|g|}$$

Levi-Civita Symbol

$$\varepsilon_{a_1 \dots a_D} \equiv \begin{cases} +1 & \text{even} \\ -1 & \text{odd} \\ 0 & \text{multiples} \end{cases}$$

$$\varepsilon^{0 \dots (D-1)} = -1$$

→ Claim: $\varepsilon_{a_1 \dots a_D}$ transforms in 1 of ISO^+

$$V'_a = \Lambda_a^b V_b$$

∴

$$\varepsilon'_{a_1 \dots a_D} = \Lambda_{a_1}^{b_1} \dots \Lambda_{a_D}^{b_D} \varepsilon_{b_1 \dots b_D}$$

$$= \varepsilon_{a_1 \dots a_D} \det(\Lambda)$$

$$= \varepsilon_{a_1 \dots a_D}$$

Since $\det(\Lambda) = 1$

□

• In general,

$$\varepsilon_{a_1 \dots a_D} \det(A) = A_{a_1}^{b_1} \dots A_{a_D}^{b_D} \varepsilon_{b_1 \dots b_D}$$

of course, for $\varepsilon_{0, \dots, D-1}$: $\det(A) = A_0^{b_0} \dots A_{D-1}^{b_{D-1}} \varepsilon_{b_0 \dots b_{D-1}}$

• Levi-Civita tensor $\text{Dens} A_\mu$:

$$e \equiv \det(e_{\mu}^a)$$

$$\xi_{\mu_1 \dots \mu_D} \equiv e^{-1} \xi_{a_1 \dots a_D} e_{\mu_1}^{a_1} \dots e_{\mu_D}^{a_D}$$

or $e \xi_{\mu_1 \dots \mu_D} = \xi \dots$ as above

$$\xi^{\mu_1 \dots \mu_D} = e \xi^{a_1 \dots a_D} e_{a_1}^{\mu_1} \dots e_{a_D}^{\mu_D}$$

are not dual and so mixed terms cannot exist.

$$\delta \xi^{\mu_1 \dots \mu_D} = 0$$

proof:

$$\delta \xi^{\mu_1 \dots \mu_D} = \delta e() + e \xi^{a_1 \dots a_D} \delta e_{a_1}^{\mu_1} \dots e_{a_D}^{\mu_D} + \dots + e \xi^{a_1 \dots a_D} e_{a_1}^{\mu_1} \dots \delta e_{a_D}^{\mu_D}$$

$$\begin{aligned} \delta e &= e \operatorname{Tr}(e^{-1} \delta e) \\ &= e e_b^M \delta e_M^b \\ &= e e_b^M \delta e_M^b \xi_{a_1 \dots a_D} e_{\mu_1}^{a_1} \dots e_{\mu_D}^{a_D} + \dots \end{aligned}$$

$$\delta e_{a_1}^{\mu_1} \dots e_{a_D}^{\mu_D} + e_{a_1}^{\mu_1} \delta e_{a_2}^{\mu_2} \dots e_{a_D}^{\mu_D}$$

What if $\delta(e_{a_1}^{\mu_1} e_{\mu_1}^b) = \delta(\delta_{a_1}^b) = 0$

$$\delta e_{a_1}^{\mu_1} e_{\mu_1}^{a_1} + e_{a_1}^{\mu_1} \delta e_{\mu_1}^{a_1} = 0$$

$$\delta e_{a_1}^{\mu_1} e_{\mu_1}^b = -e_{a_1}^{\mu_1} \delta e_{\mu_1}^b$$

$$e_b^M \delta e_{a_1}^{\mu_1} e_{\mu_1}^{a_1} = -e_b^M e_{a_1}^{\mu_1} \delta e_{\mu_1}^{a_1}$$

$$\delta e_{a_1}^{\mu_1} = -e_{a_1}^M e_b^{\mu_1} \delta e_{\mu_1}^b$$

$$(-e_{a_1}^M e_b^{\mu_1} \delta e_{\mu_1}^b) \xi^{a_1 \dots a_D} e_{a_2}^{\mu_2} \dots e_{a_D}^{\mu_D}$$

$$\delta \xi^{\mu_1 \dots \mu_D} = \delta e_{\mu_1}^b \xi^{a_1 \dots a_D} e(e_b^M e_{a_1}^{\mu_1} \dots e_{a_D}^{\mu_D} - (e_{a_1}^M e_b^{\mu_1}) e_{a_2}^{\mu_2} \dots - e_{a_1}^{\mu_1} (e_{a_2}^M e_b^{\mu_2}) e_{a_3}^{\mu_3} \dots -$$

each term: $\xi^{a_1 \dots a_D} e_{a_1}^{\mu_1} e_{a_2}^{\mu_2} \dots e_{a_D}^{\mu_D}$

$$= e_{a_1}^{\mu_1} \delta_{\sigma}^{\mu_2} \dots \delta_{\sigma}^{\mu_D} = 0 \text{ (Schooten)}$$

• First term:

$$\delta e_{\mu}^b \sum^{a_1 \dots a_D} e^{\mu} e_b^{a_1} \dots e_{a_D}^{\mu_0}$$

$$\sum^{\mu_1 \dots \mu_D} = \delta e_{\mu}^b e_b^{\mu} \sum^{\mu_1 \dots \mu_D}$$

$$\sum^{\mu_1 \dots \mu_D} (1 - \delta e_{\mu}^b e_b^{\mu}) = 0$$

$$(i) \delta e_{\mu}^b e_b^{\mu} = 1$$

$$e = e_{\mu}^a dx^{\mu}$$

$$(ii) \sum^{\mu_1 \dots \mu_D} = 0$$

→ Not sure why 1st isn't possible.

$$dV \equiv e^0 \wedge \dots \wedge e^{D-1}$$

$$\begin{aligned} dV &= \frac{1}{D!} \sum_{a_1 \dots a_D} e^{a_1} \wedge \dots \wedge e^{a_D} \\ &= -\frac{1}{D!} \sum_{a_1 \dots a_D} \sum^{1 \dots D} e^{a_1} \wedge \dots \wedge e^{a_D} \\ &= -\frac{1}{D!} (-D! \delta_{a_1 \dots a_D}^{1 \dots D}) e^{a_1} \wedge \dots \wedge e^{a_D} \\ &= \delta_{a_1}^{[1} \dots \delta_{a_D}^{D]} e^{a_1} \wedge \dots \wedge e^{a_D} \\ &= e^1 \wedge \dots \wedge e^D \end{aligned}$$

Thus,

$$\begin{aligned} dV &= e^0 \wedge \dots \wedge e^{D-1} \\ &= \frac{1}{D!} \sum_{a_0 \dots a_{D-1}} e^{a_0} \wedge \dots \wedge e^{a_{D-1}} \end{aligned}$$

Now,

$$\begin{aligned} &= \frac{1}{D!} \sum_{a_0 \dots a_{D-1}} e_{\mu_0}^{a_0} \dots e_{\mu_{D-1}}^{a_{D-1}} dx^{\mu_0} \wedge \dots \wedge dx^{\mu_{D-1}} \\ &= \frac{1}{D!} (e_{\mu_0 \dots \mu_{D-1}}) dx^{\mu_0} \wedge \dots \wedge dx^{\mu_{D-1}} \\ &= e dx^0 \dots dx^{D-1} \\ &= d^D x \sqrt{-|g|} \end{aligned}$$

$$* e^a = e_{\mu}^a dx^{\mu}$$

$$\begin{aligned} \Rightarrow de^a &= \partial_{\mu} e_{\nu}^a dx^{\mu} \wedge dx^{\nu} \\ &= \frac{1}{2} (\partial_{\mu} e_{\nu}^a - \partial_{\nu} e_{\mu}^a) dx^{\mu} \wedge dx^{\nu} \\ &= \frac{1}{2} (\partial_{\mu} e_{\nu}^a dx^{\mu} \wedge dx^{\nu}) - \frac{1}{2} \partial_{\mu} e_{\nu}^a dx^{\nu} \wedge dx^{\mu} \\ &= \frac{1}{2} (\partial_{\mu} e_{\nu}^a dx^{\mu} \wedge dx^{\nu}) - \frac{1}{2} \partial_{\nu} e_{\mu}^a dx^{\mu} \wedge dx^{\nu} \end{aligned}$$

$$de^a = \frac{1}{2} (\partial_{\mu} e_{\nu}^a - \partial_{\nu} e_{\mu}^a) dx^{\mu} \wedge dx^{\nu}$$

• de^a : transforms as a (0,2)-tensor but not as a local Lorentz vector

$$e_m^a(x) = (\Lambda^{-1})^a_b(x) e_m^b(x)$$

$$\begin{aligned} d(e^a) &= d(\Lambda^a_b(x) e^b) \\ &= \partial_\mu \Lambda^a_b e^b_\nu dx^\mu \wedge dx^\nu + \Lambda^a_b \partial_\mu e^b_\nu dx^\mu \wedge dx^\nu \\ &= (\partial_\mu \Lambda^a_b dx^\mu) \wedge (e^b_\nu dx^\nu) + \Lambda^a_b de^b \end{aligned}$$

$$de^a = d\Lambda^a_b \wedge e^b + \Lambda^a_b de^b \quad (\text{Lorentz})$$

↑
"Sprots" invariance

$$\bullet \delta(de^a - \Lambda^a_b de^b) = d\Lambda^a_b \wedge e^b$$

• Cancel two term by adding 1-form spin conn.

$$de^a + \omega^a_b \wedge e^b + \tau^a$$

when again, $\omega^a_b \rightarrow d\Lambda^a_b$

• Similar to $A_\mu \rightarrow A_\mu + \partial_\mu \Lambda$
 $\delta A_\mu = \partial_\mu \Lambda$

* Define: $\omega^a_b \equiv \Gamma^a_{cb} e^c$ *

$$\omega^a_b = \Lambda^{-1 a}_c d\Lambda^c_b + \Lambda^{-1 a}_c \omega^c_d \Lambda^d_b$$

$$\omega^a_{\mu b} = \Gamma^a_{cb} e^c_\mu$$

$$\omega^a_{\mu b} = \Lambda^{-1 a}_c \partial_\mu \Lambda^c_b + \Lambda^{-1 a}_c \omega^c_{\mu d} \Lambda^d_b$$

* Recall: $\nabla_\mu(\partial_\nu) \equiv \Gamma^\sigma_{\mu\nu} \partial_\sigma$

• Can look at this in the E_a basis

$$\nabla_a E_b \equiv \nabla_{E_a} E_b = \Gamma^c_{ab} E_c$$

$$\begin{aligned} \text{LHS } e_a^\mu \nabla_\mu (e_b^\sigma \partial_\sigma) &= e_a^\mu [(\nabla_\mu e_b^\sigma) \partial_\sigma + e_b^\sigma \nabla_\mu(\partial_\sigma)] \\ &= e_a^\mu (\nabla_\mu e_b^\sigma \partial_\sigma + e_b^\sigma \Gamma^\sigma_{\mu\lambda} \partial_\lambda) \end{aligned}$$

$$= e_a^\mu (\nabla_\mu e_b^\sigma + e_b^\lambda \Gamma_{\mu\lambda}^\sigma) \partial_\sigma$$

$$\text{RHS} = \Gamma_{ab}^c e_c^\sigma \partial_\sigma$$

$$\therefore e_a^\mu e_b^\sigma (\nabla_\mu e_b^\sigma + e_b^\lambda \Gamma_{\mu\lambda}^\sigma) = \Gamma_{ab}^c e_c^\sigma e_a^\sigma$$

$$\Gamma_{ab}^d = e_a^\mu e_b^\sigma (\nabla_\mu e_b^\sigma + e_b^\lambda \Gamma_{\mu\lambda}^\sigma)$$

$$\omega_{ab}^c = \Gamma_{cd}^a e_b^d dx^c$$

$$\omega_{ab}^c = e_a^\mu e_b^\sigma (\nabla_\mu e_b^\sigma + e_b^\lambda \Gamma_{\mu\lambda}^\sigma) e_c^\sigma$$

$$V'^a(x) = (\Lambda^{-1})^a_b(x) V^b(x)$$

$$U'_a(x) = U_b(x) \Lambda_a^b(x)$$

$$T'_{ab}(x) = t_{cd}(x) \Lambda_a^c \Lambda_b^d$$

* Lorentz: Covariant Derivatives

$$e.g.: (\Lambda^{-1})^a_b = (e^{i\xi})^a_b$$

$$\begin{cases} V'^a(x) = V^a - i(\xi V'^a) \\ \delta V'^a = -i\xi V^a \end{cases}$$

$$D_\mu V^a = \partial_\mu V^a + \omega_\mu^a_b V^b$$

$$D_\mu U_a = \partial_\mu U_a - U_b \omega_\mu^b_a = \partial_\mu U_a + \omega_\mu^b_a U_b$$

$$\omega_\mu^{ab} = \omega_\mu^a_c \mathcal{M}^{cb}, \quad \omega_{\mu a}^b = \mathcal{M}_{ac} \omega_\mu^c_d \mathcal{M}^{db}$$

$$\bullet D_\mu V^a \rightarrow D_\mu V'^a$$

$$D_\mu (\Lambda^a_b V^b) = D_\mu \Lambda^a_b V^b + \Lambda^a_b D_\mu V^b$$

RHS:

$$\partial_\mu \Lambda^a_b V^b + \Lambda^a_b \partial_\mu V^b + \omega_\mu^c_b \Lambda^a_c V^b$$

$$\partial_\mu \Lambda^a_b V^b + \Lambda^a_b \partial_\mu V^b + (\Lambda^{-1})^a_c \partial_\mu \Lambda^c_b + \Lambda^{-1}_c \omega_\mu^c_d \Lambda^d_b) \Lambda^a_b V^b$$

$$= \partial_\mu \Lambda^a_b V^b + \Lambda^a_b \partial_\mu V^b + (\Lambda^{-1})^a_c (\partial_\mu \Lambda^c_b) \Lambda^a_b V^b$$

$$\frac{1}{2} (\Lambda^{-1})^a_c \omega_m^c{}_d (\Lambda^{-1})^d_b \Lambda^b{}_f V^f$$

$$\hookrightarrow \Lambda^{-1}{}^a_c \omega_m^c{}_d V^d$$

$$\bullet (\Lambda^{-1})^b{}_f \left(-\Lambda^c{}_b \partial_m (\Lambda^{-1})^a{}_c \right) V^f$$

$$= -\delta_f^c \partial_m \Lambda^{-1}{}^a{}_c V^f = -\partial_m (\Lambda^{-1})^a{}_b V^b$$

$$= \Lambda^{-1}{}^a{}_b \partial_m V^b + (\Lambda^{-1})^a{}_c \omega_m^c{}_d \Lambda^d{}_b (\Lambda^{-1})^b{}_f V^f$$

$$= \Lambda^{-1}{}^a{}_b (\partial_m V^b + \omega_m^a{}_b V^b)$$

$$= \Lambda^{-1}{}^a{}_b (D_m V)^b$$

$$\bullet (D_m V)^a = \Lambda^{-1}{}^a{}_b (D_m V)^b \quad \checkmark$$

* Generators of $\text{Spin}(1, D-1) : \frac{i}{2} \gamma^{ab}$

* Spinor: $\text{Spin}(1, D-1) / \mathbb{Z}_2 \cong \text{ISO}^+(1, D-1)$

$$\psi'(x) = \exp\left(-\frac{i}{4} \lambda^{ab} \gamma_{ab}\right) \psi(x)$$

ex: $\int d^D x \quad \bar{\psi} \gamma^a \partial_a \psi$

$$\psi'(x) = \psi(x) - \frac{i}{4} \lambda^{ab} \gamma_{ab} \psi(x)$$

$$\delta \psi = -\frac{i}{4} \lambda^{ab} \gamma_{ab} \psi(x)$$

$$\widetilde{\delta S} = -\frac{i}{4} \int d^D x \quad \bar{\psi} \gamma^\mu \gamma_{ab} \psi \partial_\mu \lambda^{ab}$$

$$S = S_0 + \widetilde{\delta S}$$

$\uparrow \quad \partial_\mu \lambda^{ab} \rightarrow -\omega_m^{ab}$

$$S = \int d^D x \quad \bar{\psi} \gamma^\mu \partial_\mu \psi + \frac{i}{4} \int d^D x \quad \bar{\psi} \gamma^\mu \gamma_{ab} \omega_m^{ab} \psi$$

$$= \int d^D x \quad \bar{\psi} \gamma^\mu \left(\partial_\mu + \frac{i}{4} \omega_{mab} \gamma^{ab} \right) \psi$$

$\underbrace{\hspace{10em}}_{D_\mu}$

* Under Lorentz

$$(D_\mu \psi) \rightarrow e^{-\frac{i}{4} \lambda^{ab} \gamma_{ab}} (D_\mu \psi)$$

$$* \omega_{\mu[\rho\nu]} = \omega_{\mu\alpha} e^\alpha_\rho e^\beta_\nu$$

$$de^a + \omega^a_b \wedge e^b \equiv \tau^a$$

$$\left(\frac{1}{2} (\partial_\mu e^\alpha_\nu - \partial_\nu e^\alpha_\mu) + \omega^a_{\mu b} e^\alpha_\nu \right) dx^\mu \wedge dx^\nu$$

$$= \frac{1}{2} T^a_{\mu\nu} dx^\mu \wedge dx^\nu$$

$$T^a_{\mu\nu} = \partial_{[\mu} e^\alpha_{\nu]} + 2 \omega^a_{[\mu} e^\alpha_{\nu]}$$

$$T_{[\mu\nu]\rho} = e^\lambda_{\rho} g_{\lambda\rho} (2 \partial_{[\mu} e^\lambda_{\nu]} + \omega^a_{\mu b} e^\lambda_{\nu} - \omega^a_{\nu b} e^\lambda_{\mu})$$

$$= e_{\alpha\rho} (\partial_\mu e^\alpha_\nu - \partial_\nu e^\alpha_\mu)$$

$$+ e^\lambda_{\rho} g_{\lambda\rho} \omega^a_{\mu b} e^\lambda_{\nu}$$

$$- e^\lambda_{\rho} g_{\lambda\rho} \omega^a_{\nu b} e^\lambda_{\mu}$$

$$= C_{[\mu\nu]\rho} + \omega_{\mu[\rho\nu]} - \omega_{\nu[\rho\mu]}$$

$$T_{[\mu\nu]\rho} = C_{[\mu\nu]\rho} + \omega_{\mu[\rho\nu]} - \omega_{\nu[\rho\mu]}$$

Book has -

$$T_{[\mu\nu]\rho} - T_{[\nu\rho]\mu} + T_{[\rho\mu]\nu} = -2 K_{\mu[\nu\rho]}$$

$$= C_{[\mu\nu]\rho} + \omega_{\mu[\rho\nu]} - \omega_{\nu[\rho\mu]}$$

$$- C_{[\nu\rho]\mu} - \omega_{\nu[\mu\rho]} + \omega_{\rho[\mu\nu]}$$

$$+ C_{[\rho\mu]\nu} + \omega_{\rho[\nu\mu]} - \omega_{\mu[\nu\rho]}$$

$$= C_{[\mu\nu]\rho} - C_{[\nu\rho]\mu} + C_{[\rho\mu]\nu} - 2 \omega_{\mu[\nu\rho]}$$

$$K_{\mu[\nu\rho]} = \frac{1}{2} (-C_{[\mu\nu]\rho} + C_{[\nu\rho]\mu} - C_{[\rho\mu]\nu}) + \omega_{\mu[\nu\rho]}$$

$$\omega_{\mu[\nu\rho]}(e) \equiv \frac{1}{2} (C_{[\mu\nu]\rho} - C_{[\nu\rho]\mu} + C_{[\rho\mu]\nu})$$

(torsion-free spin connection)

$$W_{mnp}(e) = W_{mab} e^a e^p e^c e^d$$

$$\begin{aligned} *Calci: \frac{1}{2} & \left(\partial_m e^a_v - \partial_v e^a_m \right) e_{ap} \\ & - \left(\partial_v e^a_p - \partial_p e^a_v \right) e_{am} \\ & + \left(\partial_p e^a_m - \partial_m e^a_p \right) e_{av} \end{aligned} \quad \left(e^v_c e^p_d \right)$$

$$(i) \frac{1}{2} e^v_c \left(\partial_m e^a_v - \partial_v e^a_m \right) \underline{g_{pa} e^a e^p_d}$$

$$= \frac{1}{2} e^v_c \left(\partial_m e^a_v - \partial_v e^a_m \right) \eta_{ad} = \underline{e^v_c \partial_{[m} e^a_{v]} \eta_{ad}}$$

$$\begin{aligned} (ii) & -\frac{1}{2} e^v_c \left(\partial_v e^a_p - \partial_p e^a_v \right) e^p_d e_{am} \\ & = \frac{1}{2} e_{am} \left(\partial_p e^a_v - \partial_v e^a_p \right) e^p_d e^v_c \end{aligned}$$

$$\begin{aligned} & = \frac{1}{2} e_{am} e^v_d e^p_c \partial_v e^a_p - e_{am} \partial_v e^a_p e^p_d e^v_c \\ & = \frac{1}{2} e_{am} \partial_v e^a_p \left(e^v_d e^p_c - e^p_d e^v_c \right) \end{aligned}$$

$$= \underline{e_{am} \partial_v e^a_p e^v_{[d} e^p_{c]}}$$

$$\begin{aligned} (iii) & \frac{1}{2} e^v_c e^p_d \left(\partial_p e^a_m - \partial_m e^a_p \right) e_{av} \\ & = \eta_{ca} e^v_d - \partial_m e^a_{[v} \eta_{c]a} \\ & = -\partial_{[m} e^a_{v]} \eta_{ca} e^v_d \end{aligned}$$

$$\begin{aligned} \therefore W_{mcd} & = \partial_{[m} e^a_{v]} \left(e^v_c \eta_{ad} - e^v_d \eta_{ac} \right) \\ & + e_{am} \partial_v e^a_p e^v_{[d} e^p_{c]} \end{aligned}$$

$$W_{mcd} = 2 \partial_{[m} e^f_{v]} \eta_{f[d} e^v_{c]} + e_{fm} \partial_v e^f_p e^v_{[d} e^p_{c]}$$

$$\text{Raising by } \eta^{ca} \eta^{db}$$

$$W^{ab}_{m}(e) = 2 e^{v[a} \partial_{[m} e^{b]}_{v]} - e^{v[a} e^{b]p} e_{fm} \partial_v e^f_p$$

• Torsion-less: $\delta(de^a + \omega^a_b \wedge e^b) = 0$

$$d(\delta e^a) + \delta \omega^a_b \wedge e^b + \omega^a_b \wedge \delta e^b = 0$$

$$\partial_{[\mu} \delta e^a_{\nu]} + \delta \omega_{\mu}^a{}^b e^b_{\nu]} + \omega_{\mu}^a{}^b \delta e^b_{\nu]} = 0$$

$$\Rightarrow \boxed{D_{[\mu} \delta e^a_{\nu]} + \delta \omega_{[\mu}^a{}^b e_{\nu]}^b = 0}$$

$$e^a_\nu e^b_\rho \delta \omega_{\mu ab} = (D_{[\mu} \delta e^a_{\nu]}) e_{\rho a} - \dots + \dots$$

$$\nabla_\mu V^\nu \equiv e^\nu_a D_\mu V^a$$

$$\begin{aligned} \nabla_\mu V^\nu &= e^\nu_a D_\mu V^a = e^\nu_a (\partial_\mu V^a + \omega_\mu^a{}_b V^b) \\ &= e^\nu_a (\partial_\mu e^a_\sigma) V^\sigma + e^\nu_a e^a_\sigma \partial_\mu V^\sigma \\ &\quad + e^\nu_a \omega_\mu^a{}_b V^b \end{aligned}$$

$$= \partial_\mu V^\nu + e^\nu_a (\partial_\mu e^a_\sigma + \omega_\mu^a{}_b e^b_\sigma) V^\sigma$$

$$\nabla_\mu V^\nu = \partial_\mu V^\nu + e^\nu_a (\partial_\mu e^a_\sigma + \omega_\mu^a{}_b e^b_\sigma) V^\sigma$$

$$\boxed{\Gamma_{\mu\nu}^{\rho} = e^{\rho}_a (\partial_\mu e^a_\nu + \omega_\mu^a{}_b e^b_\nu)}$$

* Much more intuitive:

$$(e^c_\rho \cdot e^N_d) \Gamma_{\mu\nu}^{\rho} = (\Gamma_\mu)^c_d$$

$$\begin{aligned} \delta^c_a e^N_d (\partial_\mu e^a_\nu + \omega_\mu^a{}_b e^b_\nu) \\ = e^N_d \partial_\mu e^a_\nu + \omega_\mu^a{}_d \end{aligned}$$

$$\boxed{(\Gamma_\mu)^a{}_b = e^N_b \partial_\mu e^a_\nu + (\omega_\mu)^a{}_b}$$

or

$$\Gamma_{\mu\nu}^a = \partial_\mu e^a_\nu + \omega_\mu^a{}_b e^b_\nu$$

→ Not Covariant as not all 3-coords are Diff.-invariant.

$$\rightarrow p_1 \dots p_n \quad e_1 \quad p_n \quad b_1 \quad b_n \quad a_1 \dots a_n$$

$$V_\mu \Gamma_{\mu_1 \dots \mu_n}^\mu \equiv e_{a_1}^{\mu_1} \dots e_{a_n}^{\mu_n} e_{\nu_1}^\mu \dots e_{\nu_m}^\mu \Gamma_{\mu_1 \dots \mu_n}^{\nu_1 \dots \nu_m}$$

$$\star \quad \partial_\mu e_\nu^a + \omega_\mu^a{}_b e_\nu^b - e_\rho^a \Gamma_{\mu\nu}^\rho = 0 \quad \star$$

vielbein - postulate

$$\begin{aligned} \nabla_\mu (g_{\mu\rho}) &= \nabla_\mu (e_\nu^a e_\rho^b) \eta_{ab} \\ &= \nabla_\mu e_\nu^a e_\rho^b \eta_{ab} + e_\nu^a \nabla_\mu e_\rho^b \eta_{ab} \\ &= (\partial_\mu e_\nu^a + \omega_\mu^a{}_c e_\nu^c) e_\rho^b \eta_{ab} \end{aligned}$$

$$\nabla_\mu e^a = -\omega_\mu^a{}_b e^b$$

$$\begin{aligned} \nabla_\mu (e_\sigma^a dx^\sigma) &= (\nabla_\mu e_\sigma^a) dx^\sigma + e_\sigma^a \nabla_\mu (dx^\sigma) \\ &= (\nabla_\mu e_\sigma^a) dx^\sigma - e_\lambda^a \Gamma_{\mu\sigma}^\lambda dx^\sigma \end{aligned}$$

$$-\omega_\mu^a{}_b e_\sigma^b = (\nabla_\mu e_\sigma^a) - e_\lambda^a \Gamma_{\mu\sigma}^\lambda$$

$$\omega_\mu^a{}_b e_\sigma^b - e_\lambda^a \Gamma_{\mu\sigma}^\lambda + \nabla_\mu e_\sigma^a = 0$$

Comparing this with the above:

$$\cancel{\nabla_\mu e_\sigma^a} = \cancel{\partial_\mu e_\sigma^a}$$

ah... of course e is really just a function!

$$(\nabla_\mu V)^\rho \equiv e_\alpha^\rho (D_\mu V)^\alpha$$

$$\Gamma_{\mu\nu}^\rho = \underbrace{\Gamma_{\mu\nu}^\rho(y)}_{\text{L.C.}} - K_{\mu\nu}^\rho$$

* Vector-spinor field:

$$D_\mu \psi_\nu = (\partial_\mu + \frac{1}{4} \omega_\mu^{ab} \gamma_{ab}) \psi_\nu$$

$$\nabla \psi^\alpha = \eta \psi^\alpha - \Gamma_{\mu\nu}^\alpha \psi^\mu$$

$$V_M \Gamma_N = V_M \Gamma_N + \Gamma_{MN} V^P$$

* Spinors: $\nabla_\mu \psi^\alpha = D_\mu \psi^\alpha$

Define a $(\mathcal{L}, \mathcal{L})$ form

$$\tilde{\mathcal{L}} \equiv E_a \otimes e^a$$

$\tilde{\mathcal{L}} \in \text{Aut}(TM)$, since $V \in TM$

$$\begin{aligned} \tilde{\mathcal{L}}(V) &= E_a \cdot e^a_\mu dx^\mu(V) \\ &= E_a e^a_\mu V^\mu \delta^\mu_\sigma \\ &= E_a V^a = V \end{aligned}$$

i.e. $V^\sigma \partial_\sigma = V^a E_a = V$

Basis independent, so $\tilde{\mathcal{L}}$ is an identity.

along all of $TM = \bigcup_p T_p M$ since it is not dependent on the pt in the manifold. So, when parallelly transported there is no change

$$\nabla_\mu \tilde{\mathcal{L}} = 0$$

$$\tilde{\mathcal{L}} = e^a_\sigma (E_a \otimes dx^\sigma)$$

$$\nabla_\mu E_a = E_b \omega_\mu^b{}_a,$$

$$\begin{aligned} \nabla_\mu \tilde{\mathcal{L}} &= \nabla_\mu E_a \otimes e^a + E_a \otimes (dx^\sigma) \nabla_\mu (e^a_\sigma) \\ &\quad + E_a \otimes e^a_\sigma (\nabla_\mu dx^\sigma) \end{aligned}$$

$$\begin{aligned} &= (E_b \omega_\mu^b{}_a) \otimes e^a + E_a \otimes dx^\sigma (\nabla_\mu e^a_\sigma) \\ &\quad + E_a \otimes e^a_\nu (-\Gamma_{\mu\sigma}^\nu dx^\sigma) \end{aligned}$$

$$= 0$$

$$= (\omega_\mu^b{}_a e^a_\sigma + \partial_\mu e^a_\sigma - \Gamma_{\mu\sigma}^\nu e^a_\nu) (E_a \otimes dx^\sigma)$$

$$(\nabla_\mu \tilde{\mathcal{L}})_\sigma^a = \partial_\mu e^a_\sigma + \omega_\mu^b{}_a e^a_\sigma - \Gamma_{\mu\sigma}^\nu e^a_\nu$$

$$(\nabla_\mu \tilde{\mathcal{L}})_\sigma^a \equiv \nabla_\mu e^a_\sigma$$

$\therefore$

$$\nabla_\mu e^a_\sigma = \partial_\mu e^a_\sigma + \omega_\mu^b{}_a e^a_\sigma - \Gamma_{\mu\sigma}^\nu e^a_\nu = 0$$

$$* (D_\mu E)_a = E_b \omega_\mu{}^b{}_a$$

$$\nabla_\mu V^a = e^a_\mu D_\mu V^a$$

Second order Lagrangian for grav + ferm:

• Torsion free: $\omega_\mu{}^{ab}(e)$

→ Dirac field in curved spacetime

$$S = S_2 + S_{1/2} = \int d^D x e \left[\frac{1}{2\kappa^2} e^\mu_a e^\nu_b R_{\mu\nu}{}^{ab}(\omega) - \frac{1}{2} (\bar{\psi} \gamma^\mu \nabla_\mu \psi + \bar{\psi} \overleftarrow{\nabla}_\mu \gamma^\mu \psi) \right]$$

$$\bullet R_{\mu\nu\alpha\beta} = \partial_\mu \omega_{\nu\alpha\beta} - \partial_\nu \omega_{\mu\alpha\beta} + \omega_{\mu\alpha\gamma} \omega_\nu{}^\gamma{}_\beta - \omega_{\nu\alpha\gamma} \omega_\mu{}^\gamma{}_\beta$$

• Almost invariant:

$$\text{Dirac: } - \int d^D x \bar{\psi} (\gamma^\mu \partial_\mu - m) \psi$$

• Can add total divergence $\partial_\mu (\frac{1}{2} \bar{\psi} \gamma^\mu \psi)$ and leave the theory invariant.

$$S' = - \int d^D x \left[\frac{1}{2} \bar{\psi} \gamma^\mu \overleftrightarrow{\partial}_\mu \psi - m \bar{\psi} \psi \right]$$

→ $\mathcal{H}'$ is Hermitian as an operator on $\mathcal{H}$

• Use this form to generalize to curved space
Hence the above form. ($m=0$)

$$(c) \left. \frac{\delta S}{\delta \omega_{\mu ab}} \right|_{\omega=\omega(e)}$$

$$\begin{aligned} \delta S &= \int d^D x \delta e e^\mu_a e^\nu_b R_{\mu\nu}{}^{ab} \\ &+ 2 \int d^D x e \delta e^\mu_a e^\nu_b R_{\mu\nu}{}^{ab} \\ &+ \int d^D x e e^\mu_a e^\nu_b \delta R_{\mu\nu}{}^{ab} \end{aligned}$$

$\frac{\delta S}{\delta \omega_{\mu ab}}$ instead of δe , so first two vanish

$$\delta R_{\mu\nu}{}^{ab} = D_\mu \delta \omega_{\nu ab} - D_\nu \delta \omega_{\mu ab}$$

• Total derivative:

(i.e.)

$$\nabla_\mu \psi = D_\mu \psi = \partial_\mu \psi + \frac{i}{4} \omega_\mu^{ab} \gamma_{ab} \psi$$

$$\bar{\psi} \overleftarrow{\nabla}_\mu = \bar{\psi} \overleftarrow{D}_\mu = \bar{\psi} \left(\overleftarrow{\partial}_\mu - \frac{i}{4} \omega_\mu^{ab} \gamma_{ab} \right)$$

• Generators of $\text{Spin}(1, D-2)/\mathbb{Z}_2$ $v \rightarrow v^a \gamma_a$

γ^a : constant

$$\{\gamma^a, \gamma^b\} = 2\eta^{ab}$$

$$\gamma_\mu = e_{a\mu} \gamma^a, \quad \gamma^\mu = e^\mu_a \gamma^a = g^{\mu\nu} \gamma_\nu$$

• γ_μ : Lorentz - bispinor

$$(\gamma_\mu)_\alpha \xleftarrow{\text{SU}(2)} \alpha$$

$\uparrow$
Vector

$$\left(\left(-\frac{1}{4} \lambda^{ab} \gamma_{ab} \psi \right)^\alpha \right)^\dagger = \left(\delta + \frac{1}{4} (\gamma_{ab})_\alpha{}^{\dot{\alpha}} \psi^{\dot{\alpha}} \right)^\dagger$$

$$\delta \bar{\psi} = + \lambda^{ab} \bar{\psi} (\gamma_{ab})$$

$$\begin{aligned} \delta (\gamma_\mu)_\alpha{}^{\dot{\alpha}} &= -\frac{1}{4} \lambda^{ab} \gamma_{ab} \gamma_\mu + \frac{1}{4} \lambda^{ab} \gamma_\mu \gamma_{ab} \\ &= \frac{1}{4} \lambda^{ab} [\gamma_\mu, \gamma_{ab}] \\ &= -\frac{i}{4} \lambda^{ab} [\gamma_{ab}, \gamma_\mu] \end{aligned}$$

$$D_\mu \gamma_\nu = \partial_\mu \gamma_\nu + \frac{1}{4} \omega_\mu^{ab} [\gamma_{ab}, \gamma_\nu]$$

$$\nabla_\mu \gamma_\nu = \partial_\mu \gamma_\nu + \frac{i}{4} \omega_\mu^{ab} [\gamma_{ab}, \gamma_\nu] - \Gamma_{\mu\nu}^\rho \gamma_\rho$$

Claim: $\nabla_\mu \gamma_\nu = 0$

Proof: $\nabla_\mu \gamma_\nu =$

$$\partial_\mu \gamma_\nu = \partial_\mu (e_\nu^a \gamma_a) = \gamma_a \partial_\mu e_\nu^a$$

$$\frac{i}{4} \omega_\mu^{ab} [\gamma_{ab}, e_\nu^d \gamma_d] = \frac{i}{4} \omega_\mu^{ab} [\gamma_{ab}, \gamma_d] e_\nu^d$$

$$\begin{aligned} [\gamma_{ab}, \gamma_d] &= \gamma_{ab} \gamma_d - \gamma_d \gamma_{ab} \\ &= \frac{1}{2} (\gamma_a \gamma_b - \gamma_b \gamma_a) \gamma_d - \gamma_d \gamma_{ab} \end{aligned}$$

$$= \frac{1}{2} (\gamma_a (-\gamma_d \gamma_b + 2\eta_{db}) - \gamma_b (-\gamma_d \gamma_a + 2\eta_{ad})) - [\gamma]$$

$$= -\frac{1}{2} \gamma_a \gamma_d \gamma_b + \gamma_a \eta_{db} + \frac{1}{2} \gamma_b \gamma_d \gamma_a - \gamma_b \eta_{ad} - [\gamma]$$

$$= -\frac{1}{2} (-\gamma_d \gamma_a \gamma_b + 2\eta_{ad} \gamma_b) + \gamma_a \eta_{db}$$

$$+ \frac{1}{2} (-\gamma_d \gamma_b \gamma_a + 2\eta_{bd} \gamma_a) - \gamma_b \eta_{ad}$$

$$= \frac{1}{2} \gamma_d (\gamma_a \gamma_b - \gamma_b \gamma_a) + \gamma_a \eta_{db} - \gamma_b \eta_{ad} - \gamma_d \gamma_{ab}$$

$$= \gamma_a \eta_{db} - \gamma_b \eta_{ad}$$

$\therefore$

$$[\gamma_{ab}, \gamma_d] = \gamma_a \eta_{db} - \gamma_b \eta_{ad} - \eta_{ad} \gamma_b + \eta_{bd} \gamma_a$$

$$[\gamma_{ab}, \gamma_d] = 2(\gamma_a \eta_{db} - \gamma_b \eta_{ad})$$

$$\bullet \frac{1}{4} \omega_{\mu}^{ab} (2\gamma_a \eta_{db} - 2\gamma_b \eta_{ad})$$

$$= \frac{1}{2} (\omega_{\mu}^{ab} \gamma_a \eta_{db} - \gamma_a \eta_{bd} \omega_{\mu}^{ba}) e_{\nu}^d$$

$$= \omega_{\mu}^{ab} \gamma_a \eta_{db} e_{\nu}^d$$

$$= \omega_{\mu}^{ab} \gamma_a e_{b\nu}$$

$$\Rightarrow \nabla_{\mu} \gamma_{\nu} = \gamma_a \partial_{\mu} e_{\nu}^a + \gamma_a \omega_{\mu}^{ab} e_{b\nu} - \Gamma_{\mu\nu}^{\rho} e_{\rho}^a \gamma_a$$

$$= \gamma_a (\partial_{\mu} e_{\nu}^a + \omega_{\mu}^{ab} e_{b\nu} - \Gamma_{\mu\nu}^{\rho} e_{\rho}^a)$$

$$= 0 \quad (\text{vielbein postulate}) \quad \checkmark$$